3. The following **Table WP-01-Supplement** provides brief information — including the objectives and scope of work — for the projects listed in Table WP-01 that require funding support from the Commission. Complete details on the terms of reference for each project are available at SC21-GN-WP-04 (*Terms of reference for proposed projects for 2026-2028*).

TABLE WP-01-supplement

II. Priority ranking - Required							
No.	**Project Title**	**2026**	**2027**	**2028**	**Notes**	**Data**	**Score**
10	**P68.** Seabird mortality	30,000			Responsibility: SPC	N/A	**3.6**

Objectives

- Fulfil the requirement under the WCPFC seabird CMMs to estimate the total number of seabirds being killed per year in WCPFC fisheries.
- Assess mortality per year over the ten years since the first WCPFC seabird CMM, as requested under CMM2006-02, CMM 2007-04 and CMM 2012-07, and assess whether there is any detectable trend.
- Provide advice to the Scientific Committee on what data improvements are needed to enable better analyses to be made.

Scope of work

- Identify the limitations in the data available.
  - Document what information is available on species that overlap with the WCPFC fisheries, their population status, any tracking data (or already established spatial distributions that are more complex than binary presence absence) and any estimates on their biological parameters such as Rmax (the theoretical maximum breeding rate).
  - Document current observer data to assess the level of coverage in each fishery, both spatially and temporally.
- Estimate species-specific seabird mortalities (where feasible) and risks associated with interactions with WCPFC fisheries (where feasible).
- Describe the methods used to estimate total mortalities, including treatment of data gaps.
- Provide advice to the Scientific Committee on the suitability of risk assessment approaches given the data available.
- Generate advice on what further level of seabird assessment at the species level can be conducted, given the amount and quality of data currently available.

11	**100d.** Application of CKMR methods to SP albacore and cost-benefit evaluation and prioritisation of CKMR to WCPFC stocks (tuna, billfish, sharks)	250,000	250,000		Responsibility: SPC (New project)	No	**6.3**

Objectives

- Provide CKMR estimated demographic parameters for use in the 2027 south Pacific Albacore stock assessment.
- Provide guidance to WCPFC-SC on future CKMR applications to WCPFC stocks.

Scope of work

(i) A further 50,000 tissue samples of south Pacific albacore are required to meet the 84,000 maximum estimated by SC20-SA-IP-24. Genotyping of up to 69,000 samples is required to identify the number of kin for CKMR estimation with a CV of 15% (ie . 15,000 have already been genotyped by P100c). Bioinformatics for the genotyped samples is required to identify kin which will then be embedded into a population dynamics model and used to estimate absolute adult abundance, mortality rates, and connectivity. The CKMR population dynamics model will need to be adapted to fit the life-history of south Pacific albacore. The genotyping will also allow conformation of the number of genetic groups (see SC21-SA-10).

(ii) The second part of P100c continuance requires a desktop evaluation of future potential CKMR application. The study will draw on the global experience to provide WCPFC-SC with a guide on the approximate number of samples that need to be collected per WCPFC stock, the number of years of collection needed to obtain a first estimate, the logistical issues with collection, any genomic assay development needed, and expected costs to achieve the first estimate and then subsequent estimates.

12	**P121.** Ecosystem and Climate Indicators	15,000	15,000		Responsibility: SPC (Ongoing)	N/A	**6.3**

Objectives

- Develop and test candidate ecosystem and climate indicators to track the impact of climate and ecosystem changes on WCPFC fisheries and ecosystems.
- Provide technical advice to the Scientific Committee on the suitability of criteria used for testing and evaluating the performance of candidate indicators.
- Support the Scientific Committee in developing tools to communicate ecosystem and climate change impacts to WCPFC and external stakeholders and interest group.

Scope of work

- Technical analyses to develop and test candidate indicators.
- WCPFC member and expert workshops to refine indicators.
- Scientific Committee Reporting.
- Routine preparation of adopted indicators
- Development of tools for communication to WCPFC and wider stakeholders

13	**P122a**. Extending the scoping study on longline effort creep in the WCPO to a broader longline CPUE project: support a cross-tuna RFMO collaborative technical workshop on longline CPUE abundance index methods, issues, and good practices	20,000			Responsibility: SPC (Ongoing)	N/A	**7.1**

Objectives

1. Conduct a comprehensive stocktake and synthesis of longline CPUE standardization approaches across all five tuna RFMOs.
2. Develop and publish technical guidelines specific to tuna longline CPUE analysis, addressing identified issues and gaps in current methods.
3. Provide actionable recommendations for each tuna RFMO to improve CPUE standardization for bigeye, yellowfin, albacore and bluefin tuna assessments.
4. Produce a set of technical peer reviewed papers exploring key issues with tuna longline CPUE analysis and recommended good practices to support best available science on longline abundance indices for tuna RFMOs.
5. Establish ongoing collaboration mechanisms between tuna RFMO CPUE analysts to facilitate knowledge sharing and methodological consistency.

Scope of work

This project will support:

1. Convening a 5-day technical workshop on longline CPUE analysis attended by analysts involved in CPUE modelling across all tuna RFMOs, with invited experts in relevant statistical modelling methods.
2. The workshop would cover the key themes:
   - Fishery structure and data
   - Analytical methods
   - Implementation and application in stock assessment
3. The outcomes of the workshop would include:
   - Peer-reviewed papers compiled for a special theme/issue in Fisheries Research
   - Synthesis paper reviewing current practices across all tuna RFMOs
   - Good practices guidelines specific to tuna longline fisheries CPUE analysis
   - Workshop report with actionable recommendations by RFMO

14	**P123**. Scoping the next generation of tuna stock assessment software	50,000			Responsibility: SPC (Ongoing)	Yes	**7.3**

Objectives

- Ensure that WCPFC stock assessments are of the highest possible quality of science, forming the basis of advice for sustainable management of tuna, billfish, and shark stocks.

Scope of work

- The work on the SPC-IATTC tuna software development will focus on the initial software development design, following the outcomes from the CAPAM online workshop in December 2025. This initial design work is expected to deliver functional prototype components (core model objects) containing the population, observed data, and model predictions, depending on resources and the outcomes of the December workshop. A software repository containing source code, examples, and documentation, describing the functional prototype components of the software project will be produced in 2026.
- The work on the analysis of WCPO skipjack tagging data will focus on producing abundance indices and a corresponding write-up of the methods, results and utility for stock assessments. The deliverables will be a repository containing the analysis and a short technical report describing the analysis for SC22.
- A proposal will be developed for SC22 describing the next phase of work.

15	**P127a.** Additional resources to P127 (Reconciliation of size composition data for stock assessments)	50,000			Responsibility: SPC (Ongoing)	Yes	**6.6**

Objectives

Additional resources for:

1. Conduct a detailed analysis of current size data coverage, initial focus on yellowfin and bigeye tuna (20K USD)
2. Targeted conversion factor sampling – fund Sokskargen annual sampling, and observer/port sample tasking (20K USD)

Scope of work

1. Conduct a targeted statistical and mapping analysis to identify patterns of size data collection among fleets and model regions and identify imbalances in size data in relation spatial and fleet catches. It will provide recommendations for targeted improvements for size data collection efforts, and a basis for developing more refined statistical methods for size data standardization. This work may focus on bigeye and yellowfin, and the methods developed could be applied to other tuna and billfish as part of routine assessment developments.
2. This component will build on the conversion factors review conducted and reported in ST-WP-02, and conduct target sampling on problematic conversion factors, including the neglected length-length and weight-weight conversions. We propose to continue to support the Sokskargen annual sampling program and target observer tasking to collect conversion factor data.

16	**P128a**. Initial analyses to support investigations of the connectivity of key tuna species between the Western Pacific and East Asia (WPEA) region and the broader WCPFC-CA	125,000			Responsibility: CSIRO (New project)	Yes	**6.3**

Objectives

- The aim of this project is to undertake analyses to: 1) understand the influence of different connectivity scenarios for YFT, and possibly SKJ, between the WPEA region and the broader WCPFC-CA on regional stock assessments in the WCPO and 2) assess the feasibility of using a novel genomic approach – Low Coverage Whole Genome Sequencing (LCWGS) – for assessing connectivity (including minimum sample size and analytical requirements).
- The specific objectives are to:
  - i. Undertake modelling to assess the impact of different connectivity assumptions between the WPEA region and broader WCPFC-CA on the results of regional stock assessments for YFT (and possibly SKJ).
  - **ii.** Undertake a preliminary examination of the utility and sample size requirements of using LCWGS for investigating connectivity of YFT as a test species within the WPEA region and adjacent waters using existing genetic samples.

Scope of work

The proposed activities include:

(i) Modelling to assess the impact of different connectivity hypotheses on the results of the regional YFT (and potentially SKJ) stock assessments and subsequent management advice.

(ii) LCWGS sequencing and analysis of resulting data from a subset of existing YFT samples, collected at key locations during ACIAR project FIS/2009/059, using LCWGS (up to 50 samples from each of Maldives, Palabuhanratu (IDN), Kendari (IDN), Ambon (IDN), Bitung (IDN), Jayapura (IDN), and Noro (SLB)).

(iii) An in-person stakeholder workshop involving participants from each of IDN, PHL, VNM, SPC, and CSIRO to discuss results of (i) and (ii) above and progress full project proposal development.